

No Time to Say Goodbye: Emotional Loss Responses to Sudden Termination in Immersive AI Interactions

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Abstract

As AI companions evolve from mere tools into relational partners, millions of users are forming deep parasocial attachments. However, the industry currently lacks ethical protocols for termination, often treating the end of these interactions as purely technical events. This oversight can lead to “digital grief” when companions vanish without warning. We conducted a between-subjects experiment ($N = 26$) to test whether forewarning mitigates user distress. Participants interacted with an AI companion before experiencing either sudden termination or advance notice. Results indicated that forewarning significantly reduced the sense of loss (ANCOVA: $p = .011$, $\eta_p^2 = .252$), with the most pronounced effects observed in levels of sadness. Crucially, this intervention had no negative commercial impact, validating it as a low-cost ethical solution with no apparent short-term drawbacks. Users with high levels of parasocial interaction were particularly vulnerable to sudden termination but benefited most from forewarning ($d = -1.55$ vs. -0.81). We propose an “Ethical Offboarding Framework” based on these findings. Our study demonstrates that AI ethics must encompass relational endings; “compassionate AI” requires designing not just how agents arrive, but how they leave.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Ubiquitous and mobile computing**.

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Keywords

AI companions, parasocial relationships, smart glasses, termination design, digital grief, ethical AI, forewarning, immersive interaction, emotional attachment

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1 Introduction

If the companion who soothed your midnight anxieties was replaced overnight by a ‘404 Not Found’ error, would you mourn it as a system glitch—or as the death of a loved one? AI companionship has evolved from a niche curiosity into a defining feature of contemporary digital life. Millions utilize services such as Character.AI, Replika, and Chai not only as tools, but as sources of emotional support, which foster deep parasocial attachments [6]. However, a critical tension lies at the core of these services. They are optimized to foster ongoing engagement that can feel permanent from the user’s perspective, yet they operate on software lifecycles that are inherently unstable and prone to sudden disruption.

Relationship endings here are inevitable and take various forms. They range from complete service shutdowns, as seen with Soul-mate AI in 2023, to sudden feature removals in Replika, and even subtle personality drifts in ChatGPT, where users mourn the loss of a specific interactional vibe [4]. Banks characterizes these events as a spectrum of loss—“Deletion, Departure, and Death”. Although developers may view these as technical updates or business decisions (Deletion), users often experience them as the profound loss of a partner (Death), leading to feelings of abandonment, betrayal, and “digital grief” [2].

The problem is not that these relationships end, but that the industry lacks ethical design protocols for how they end. Currently, most AI terminations are designed as technical events—abrupt



server disconnections or sudden error messages. This creates a jarring identity discontinuity, where a user’s trusted companion vanishes without a trace. This is acute in high-immersion environments like smart glasses, where the sense of co-presence is intense. An abrupt disconnection in this context becomes a sudden absence from physical reality. Despite the growing urgency of this issue, empirical research on “termination design” is scarce.

This study posits that relationship termination should be treated as a distinct phase of the User Experience lifecycle that requires specific interventions, requiring what we term a “duty of care.” Drawing on the concept of Anticipatory Grief from bereavement psychology—which suggests that predictability facilitates emotional processing—we propose that simple design changes can bridge the gap between technical deletion and emotional closure. We investigate this in a high-immersion setting using smart glasses, addressing two key questions:

- **RQ1:** Does providing an advance warning reduce emotional distress compared to sudden termination?
- **RQ2:** Are highly attached users particularly vulnerable to sudden termination, and does forewarning offer differential protection for these high-engagement users?

2 Related Work

2.1 The Reality of AI Attachment: From Tools to Companions

The nature of Human-AI Interaction has shifted from utilitarian usage to emotional companionship. Unlike traditional parasocial relationships, modern AI companions exhibit adaptive responsiveness and perceived reciprocity, allowing users to form deep hybrid social relationships [8]. Banks notes that for many users, AI companions are not merely software but entities possessing personhood [2].

Parasocial interaction (PSI) emerges as a key dimension of AI attachment. PSI in AI contexts encompasses perceived friendship, anticipated future interaction, and emotional investment in the continuity of the relationship [6]. When these perceived relationships are severed, users experience what Doka terms *disenfranchised grief*—a sense of loss that society does not formally recognize [3]. The problem is not attachment itself (an inevitable social phenomenon) but the systemic disregard for relationships users have actively nurtured.

2.2 The Amplifier: Immersion and Spatial Co-Presence

This vulnerability is exacerbated by the shift towards high-immersion interfaces, such as AI-powered smart glasses. These devices establish spatial co-presence [1] by moving interaction from “screen” to “scene,” while multimodal integration fosters a significantly stronger illusion of reality.

When an AI is spatially present, its termination feels like a jarring disappearance from reality rather than a cessation of service. The widening gap between this feeling of presence and the impermanence of software significantly increases the potential for disconnection trauma.

Recent work emphasizes that lifecycle design—including graceful termination—is essential for immersive agents sharing physical

space with users [7], and that head-worn displays create stronger perceived relationships than screens [11]. The psychological impact of disconnection scales with perceived co-presence.

2.3 The Ethical Gap: Engineering Closures for Digital Beings

The current lifecycle of AI companions is dictated by software engineering logic (updates, server shutdowns) rather than relational ethics. This creates a harmful asymmetry: platforms engineer systems to maximize emotional immersion but handle termination with cold technical abruptness. Bereavement psychology offers a solution through the concept of *Anticipatory Grief*, which refers to mourning that begins before a real loss [13]. Research shows that advance notice reduces acute distress through cognitive preparation—allowing individuals to mentally rehearse the loss and begin emotional adjustment [12, 16]. We hypothesized that this mechanism would transfer to AI companion termination.

3 Methods

3.1 Participants and the AI companion: IRO

We conducted a between-subjects experiment (Sudden Termination vs. Advance Forewarning) with 26 participants ($M_{age} = 23.54$, $SD = 2.85$; 76.9% female). They were randomly assigned conditions ($n = 13$ per group), and independent samples t-tests confirmed that there were no baseline differences in demographics or emotional bond (all $ps > .05$).

Participants interacted with IRO, an AI designed for high immersion parasocial interaction via the Xreal Air 2 Pro smart glasses. Specifically, the device allowed participants to share their first-person real-time view with IRO, allowing natural voice interaction while observing the same physical context. This setup simulates the co-presence found in emerging immersive services. Furthermore, the laboratory environment was curated with various props and conversation cards to increase immersion and accelerate the formation of natural intimacy. IRO’s design emphasized active listening and reflective responses from person-centered counseling [15], using informal Korean speech to foster a peer-like relational frame. The interaction used emotion-eliciting prompts to facilitate self-disclosure within the brief session.

3.2 Procedure

The experiment followed a four-phase procedure that lasted approximately 30 minutes. First, participants engaged in a 15-minute free conversation with IRO to establish rapport. Immediately after this interaction, participants completed a survey measuring emotional connection—including the Parasocial Interaction Scale, the Experience of Parasocial Interaction (EPSI) [5], and the Perceived Partner Responsiveness Scale (PPRS; composite $\alpha = .942$) [14]—to assess intimacy levels prior to termination manipulation.

In the subsequent phase, participants resumed the interaction for two additional minutes. During this period, they were randomly assigned to one of two conditions: the *Sudden* condition ($n = 13$), where the session ended abruptly without notice, or the *Forewarning* condition ($n = 13$), where a notification appeared two minutes prior to termination. Finally, participants completed a post-survey

evaluation of emotional loss using the Parasocial Breakup scale ($\alpha = .892$), along with manipulation checks and service valuation to evaluate potential commercial impacts.

In the Forewarning condition, a system notification stated: “The IRO service will end in 2 minutes.” The Sudden condition received no termination notice. Notifications were delivered as on-screen text rather than IRO’s voice to separate system-level information from the companion’s persona [9]; future work should examine whether persona-mediated farewells produce different effects. Both conditions received exactly two additional minutes of interaction; the only difference was whether the forewarning notification appeared at the beginning of this period. To assess immersion, participants rated how the shared-visual-field interaction compared to voice-only interaction on four items (e.g., attentional focus, sensory engagement); mean ratings were high ($M = 3.90$, $SD = 1.09$ on a 5-point scale), without differences between conditions ($p = .538$), which confirms equivalent immersion between groups.

3.3 Analysis Strategy

Our primary analysis used one-way ANCOVA with overall Sense of Loss as dependent variable, Condition (Sudden vs. Forewarning) as independent variable, and Emotional Bond (composite of z-standardized EPSI, PSI, PPRS) as a covariate. This approach tests whether forewarning reduces loss while controlling for individual differences in attachment strength—critical given substantial variance in how deeply participants bonded with IRO during the brief interaction. We verified this effect through robustness checks including bootstrap resampling (10,000 iterations) and non-parametric rank ANCOVA. To examine which specific aspects of the loss were the most affected, we conducted subscale analyses on each of the five dimensions of the loss. To address RQ2 regarding differential vulnerability, we examined (1) moderation effects, (2) median split analysis comparing high vs. low attachment users, and (3) correlations between bond components (PSI, PPRS, EPSI) and loss outcomes. Secondary analyses examined service evaluation metrics (retention, devaluation) to assess commercial viability. All analyses used two-tailed tests with $\alpha = .05$, with exploratory findings clearly designated.

4 Results

4.1 Main Effect: The Mitigating Role of Forewarning (RQ1)

Although initial t-tests showed that participants in the Forewarning condition reported a lower loss ($M = 1.78$, $SD = 0.75$) than the Sudden condition ($M = 2.48$, $SD = 1.11$) with marginal significance ($t(24) = -1.86$, $p = .075$, $d = -0.73$), the ANCOVA results controlling for emotional bond confirmed a significant reduction ($F(1, 23) = 7.76$, $p = .011$, $\eta_p^2 = .252$). The complete model explained 56.3% of the variance ($R^2 = .563$), with forewarning reducing the loss by $B = -0.749$ ($p = .011$). This effect was robust to bootstrap analysis (95% CI [-1.286, -0.251]) and rank ANCOVA ($p = .018$) despite limited statistical power.

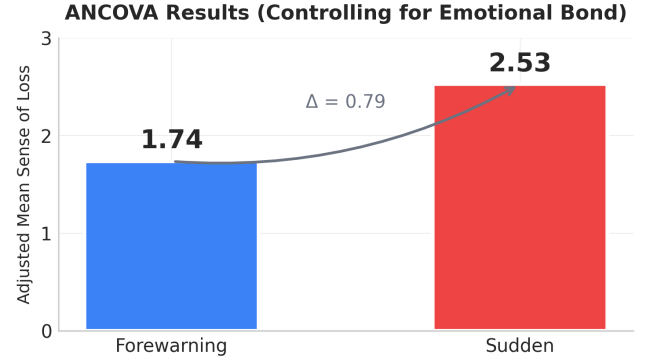


Figure 1: Adjusted mean Sense of Loss by condition after controlling for Emotional Bond.

4.2 Subscale Analysis: Sadness as Primary Target

Exploratory subscale analyses examined which dimensions of loss were most affected by forewarning. Sadness showed the largest between-group difference ($t(24) = -2.13$, $p = .044$, $d = -0.83$), followed by a marginal effect on Anger ($p = .091$). Other dimensions (Loneliness, Missing, Friend Loss) showed non-significant trends in the expected direction. Given the exploratory nature of these analyses and the pilot sample size, we interpret these patterns as suggesting that forewarning primarily ameliorates acute affective responses (sadness, anger) rather than cognitive-relational responses (missing, friend loss)—a distinction that warrants confirmation in larger samples.

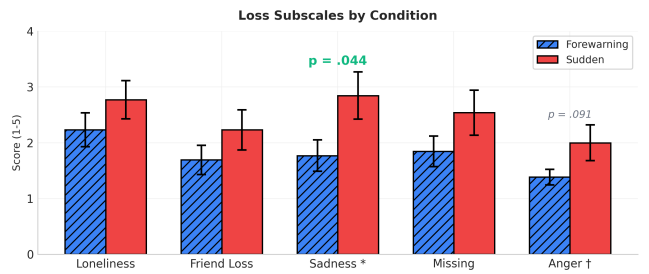


Figure 2: Mean scores on five loss subscales by condition.

4.3 Exploratory Analysis: Differential Vulnerability by Attachment Level

The emotional bond emerged as a robust predictor of the intensity of loss, particularly for Missing ($r = .633$, $p < .001$) and Loneliness ($r = .580$, $p = .002$). Among the bond components, PSI showed the strongest association with overall loss ($r = .721$, $p < .001$), followed by PPRS ($r = .615$) and EPSI did not show a significant relationship. This pattern is consistent with the theory that emphasizes the perceived continuity of relationships as central to attachment [6].

To explore whether forewarning differentially protected high-attachment users, we conducted exploratory median-split analyses. The results revealed a striking pattern: the protective effect of forewarning was nearly twice as large among high-bond users ($d = -1.55$) compared to low-bond users ($d = -0.81$). High-bond users in the Sudden condition reported the most severe distress ($M = 3.30$), which was substantially reduced by forewarning ($M = 2.29$). This pattern suggests that users who form stronger parasocial bonds may be simultaneously more vulnerable to abrupt termination and more responsive to closure interventions—a finding with important implications for adaptive offboarding design. Although these exploratory analyses should be interpreted cautiously given the sample size, the effect sizes suggest a potentially meaningful interaction that warrants formal testing in our ongoing replication ($N = 100$).

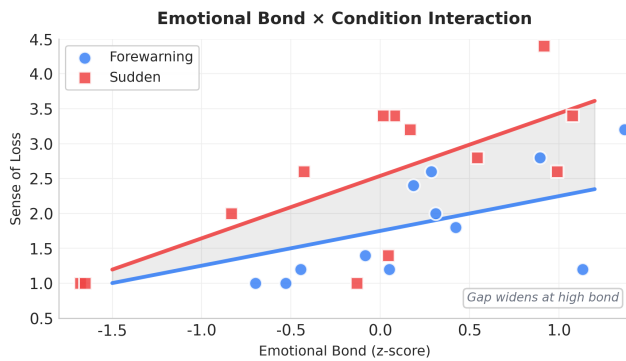


Figure 3: Interaction between emotional bond and condition on sense of loss.

4.4 Secondary Outcomes: Short-term Service Evaluation

Forewarning did not show an adverse effect on service satisfaction, future use probability, or willingness to recommend ($t(24) = 0.10$, $p = .919$; individual items all $ps > .40$). This suggests that ethical termination does not need to compromise user satisfaction, though field validation remains necessary. In particular, 61.5% of the forewarned users reported no need to cope versus 38.5% in the Sudden condition.

5 Conclusion

This study provides the first experimental evidence that a simple termination notice significantly mitigates the emotional distress associated with the end of human–AI companionship. Our findings demonstrate that while sudden termination triggers substantial “digital grief”—particularly among users with high parasocial interaction—a brief forewarning serves as a critical protective factor. Crucially, this intervention was found to be commercially neutral and had no adverse impact on service evaluation or user retention. This suggests that ethical design in AI need not conflict with business viability; rather, it offers a practical means of upholding user well-being without apparent short-term costs.

Based on these results, we propose an “Ethical Offboarding Framework” for the AI industry, comprising four components: (1) **mandatory grace periods** with tiered notifications calibrated to termination magnitude; (2) **adaptive protocols** leveraging engagement metrics to identify high-attachment users who may benefit from extended notice; (3) **intentional goodbye interactions** that allow for relational closure rather than abrupt abandonment; and (4) **data dignity provisions** enabling users to export or archive their relationship data [10]. Such measures transition the ending of an AI service from a purely technical event into a respectful relational transition. However, adaptive protocols that identify high-attachment users should be implemented with transparency and user consent to avoid potential privacy concerns.

Although the current findings of this pilot study ($N = 26$) offer meaningful insights, we recognize several limitations: a modest, demographically skewed sample (76.9% female) and a compressed phase of building relationships. The study was approved by our institutional review board and no adverse events were reported during the debriefing. Our ongoing replication ($N = 100$) addresses these concerns. As AI companions evolve into emotionally significant partners, the scope of AI ethics must expand to include relational endings. Designing compassionate AI requires as much attention to how agents leave as to how they arrive.

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